

Motzkin Neighborhood Evaluation in Iterated Local Search and Simulated Annealing for the Permutational Flow Shop Problem

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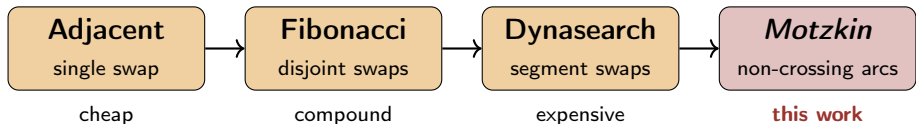
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Introduction

Permutation Flow Shop Problem (PFSP) — n jobs, m machines, same order, minimize $C_{\max}$. NP-hard for $m \geq 3$.

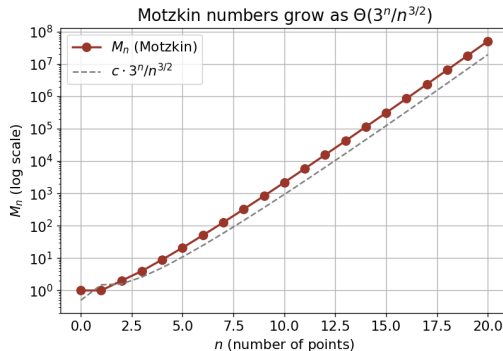
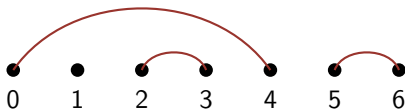
Four classical neighborhoods compared inside ILS and SA:



Motzkin Numbers and Neighborhood Structure

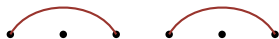
Motzkin numbers M_n count *non-crossing* arc configurations on n ordered points (isolated points allowed).

Example on 7 points — arcs
 $(0, 4), (2, 3), (5, 6)$:



Admissibility Rules

A composite move $\mathcal{M} = \{(i_1, j_1), \dots, (i_k, j_k)\}$ is admissible iff:



✓ disjoint



✗ crossing



✓ nested

Plus **index disjointness**: $\{i_a, j_a\} \cap \{i_b, j_b\} = \emptyset$.

Each (i, j) pair denotes an *end swap* of segment $[\pi_i \dots \pi_j]$: only the endpoints exchange, the interior stays untouched.

Constructing the Best Composite Move

Single-iteration optimization via dynamic programming:

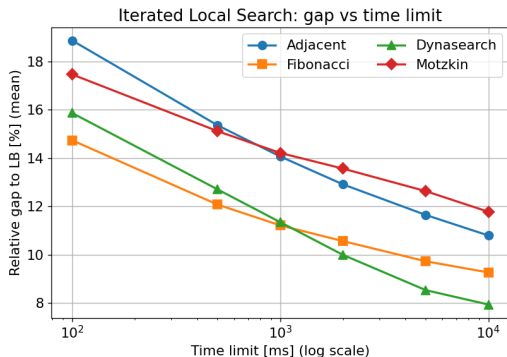
- 1 Compute Head and Tail completion-time matrices — $O(mn)$.
- 2 Evaluate all pairwise end-swap deltas $\Delta_{i,j}$ via Head+Tail — $O(mn^2)$.
- 3 DP picks optimal set of non-crossing, non-overlapping pairs — $O(n^3)$.
- 4 Apply selected swaps; recompute $C_{\max}$.

Trade-off: one Motzkin iteration explores up to M_n configurations in $O(n^3)$ time — fewer rich moves vs. many simple ones (Adjacent: $n-1$ swaps in $O(mn)$).

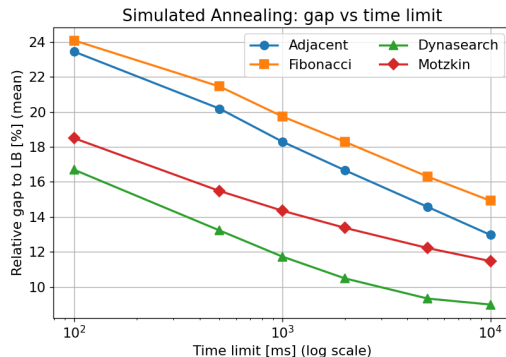
Experimental Setup

- Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$ — 12 instance classes.
- Time limits: $t_{\max} \in \{100, 500, 1000, 2000, 5000, 10000\}$ ms.
- ILS: tabu list $\tau_{\text{tabu}} = 10$.
- SA: $T_0 = 1000$, $T_{\min} = 1$, $\alpha = 0.95$, reheat $r = 1.5$, stagnation 500 ms.
- Compared neighborhoods: Adjacent, Fibonacci, Dynasearch, *Motzkin*.
- Metric: mean RPD to lower bound.

Results — Gap vs. Time Limit



ILS



SA

Dynasearch lowest gap; *Motzkin* beats Adjacent at larger $t_{\max}$; Fibonacci converges fastest but plateaus highest.

Statistical Significance (Wilcoxon signed-rank)

Dynasearch

vs. all others

$p < 0.01$

all $t_{\max}$

Motzkin

vs. Fibonacci

$p < 0.05$

$$t_{\max} \geq 5 \text{ s}$$

Motzkin

vs. Adjacent

n.s.

Competing effects: Motzkin runs few rich moves; Adjacent runs many simple ones. Their per-iteration gains tend to offset over the 12 instance classes (limited sample size).

Conclusion and Future Work

Summary:

- Motzkin: structured multi-swap neighborhood based on non-crossing arcs.
- DP enables $O(n^3)$ exploration of an exponentially large move set.
- Beats Adjacent at larger time budgets in both ILS and SA.
- Falls behind Dynasearch, which has a richer candidate set at lower cost.

Future work:

- Improve computational efficiency to allow more iterations per budget.
- Quantum-assisted Motzkin evaluation (D-Wave QUBO, QAOA).
- Hybrid neighborhood strategies — cyclic / adaptive selection.